

The Geography of Dining: Neighborhood Income and Restaurants in Chicago

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Project Description

Map Chicago's "geography of dining" by linking restaurant features to neighborhood socioeconomic status (SES)

- SES measure: median household income (community area)
- Test whether higher-SES neighborhoods differ in
 - restaurant prices (price level)
 - broader characteristics such as restaurant type, service options (takeout, delivery, dine-in, reservable), opening-hours, ratings, and review volume

RQ1: How are restaurant prices (price level) associated with neighborhood socioeconomic status (SES), measured by median household income, across Chicago?

- H1: Higher-SES neighborhoods → higher average price levels + a greater concentration of expensive restaurants

RQ2: Do restaurant characteristics (type, service options, hours, ratings, and review count) vary systematically across neighborhoods with different socioeconomic status?

- H2: Higher-SES neighborhoods will have:
 - More full-service and reservable restaurants
 - Greater diversity of restaurant types
 - Higher average ratings and review counts

Why it matters:

- Evidence of spatial inequality
- Reflects neighborhood stratification and commercial development
- Reveals how consumption patterns shape urban space

Data

Source	Sample Size	Time Period	Collection Method	Key Variables Created
Google Maps	3,388 → 1,968 matched restaurants	2026	Google Places API	price_level, rating, review_count, service flags (takeout, delivery, dine-in, reservable), primary_type
Yelp	5,112 → 1,968 matched restaurants	2026	Yelp Fusion API	price_level, rating, review_count, service flags (takeout, delivery, dine-in, reservable), primary_type
ACS 5-Year Data (Chicago Data Portal)	77 community areas	2018–2022	Download	median_household_income
Chicago Community Area Boundaries (Chicago Data Portal)	77 polygons	current shapefile	Download	spatial join to assign restaurants to community areas

Lessons Learned and Improvements

Challenge:

- Limited API calls forced tradeoffs between coverage and depth
- Nearby Search returns a maximum of 20 results, which can bias spatial coverage



scraping around one point



ZIP 60637



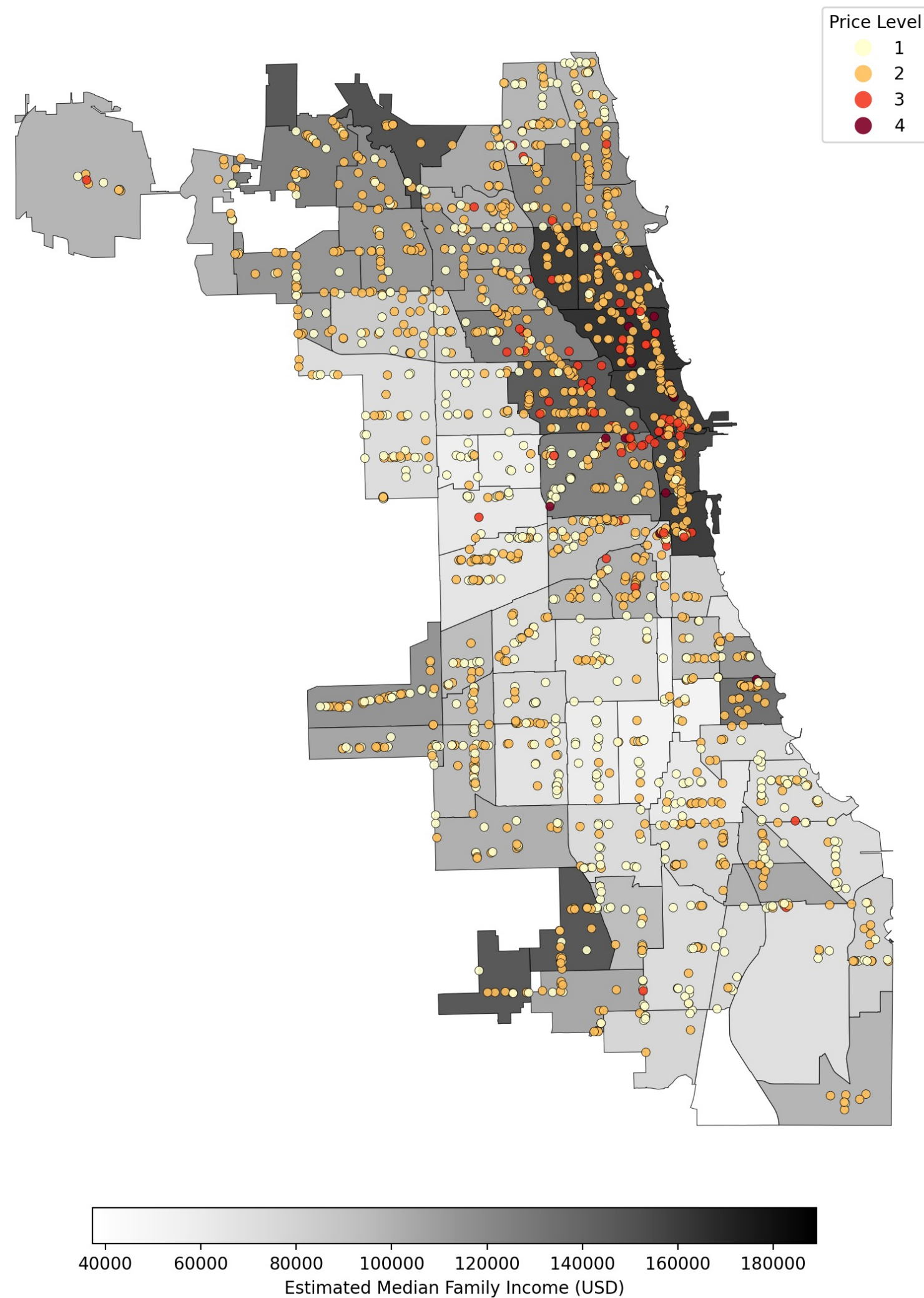
citywide grid sampling inside the Chicago boundary polygon



Plan sampling strategy before scraping

- Generate grid points
- Run nearby search around each grid point
- Deduplicate by restaurant id
- Adjust grid step size

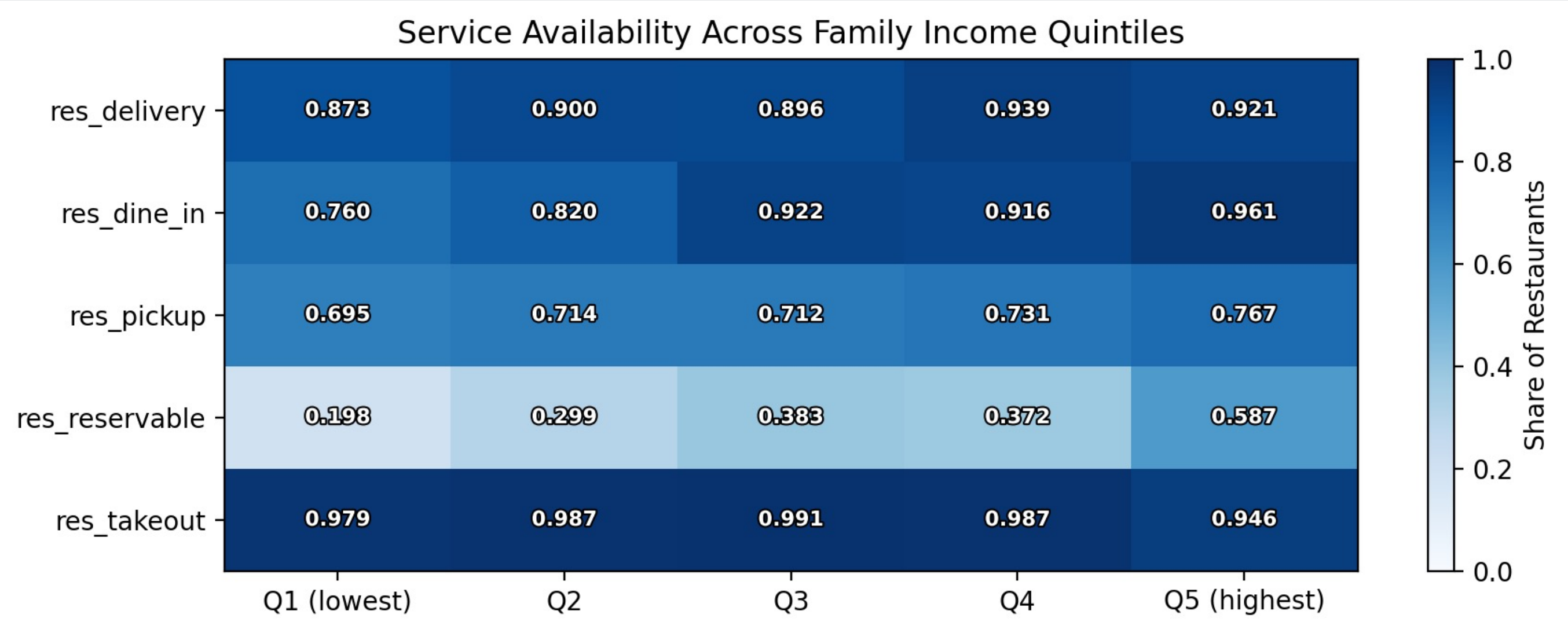
Spatial Correlation: Restaurant Price Levels & Community Income



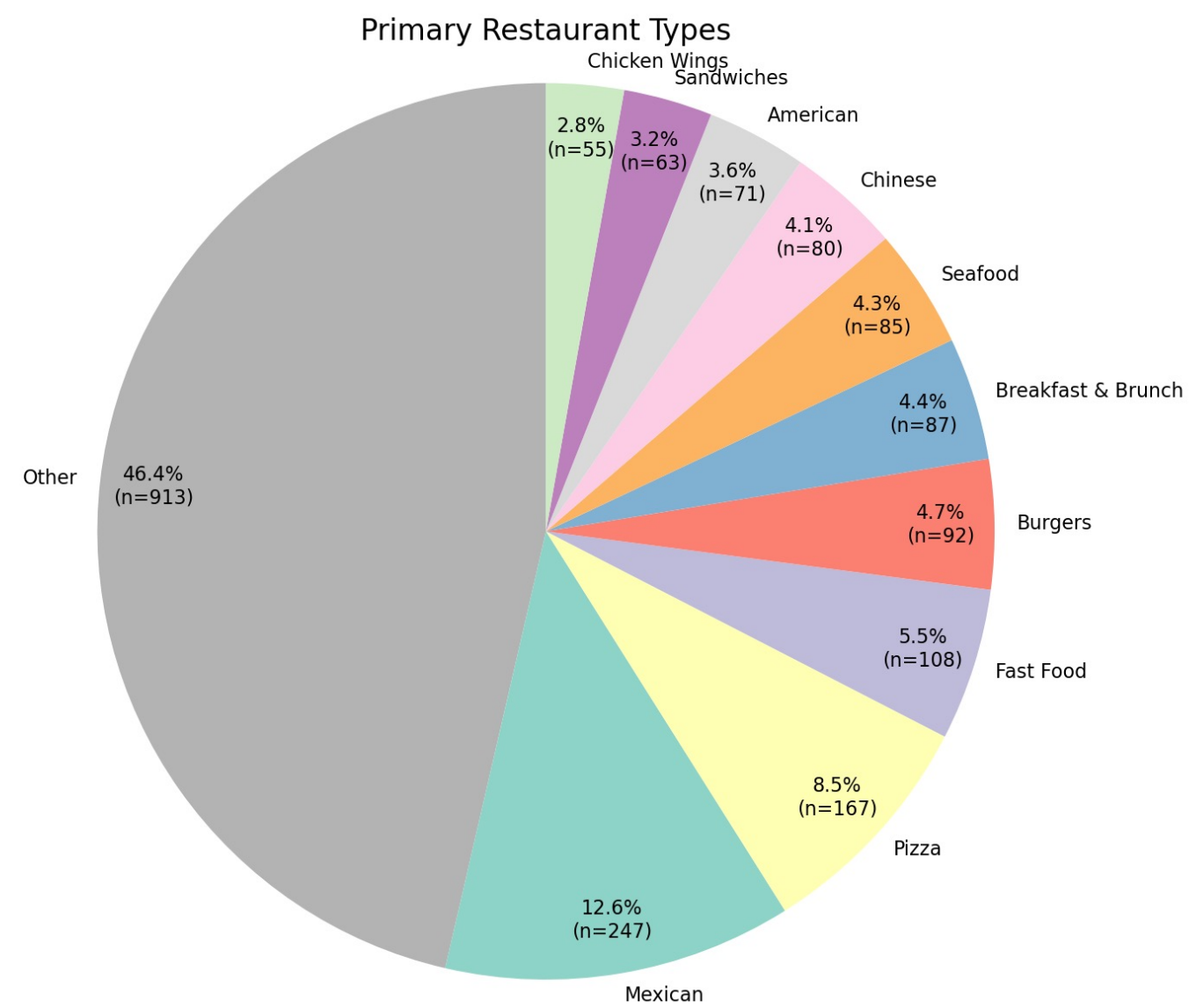
Spatial Correlation: Restaurant Price Levels & Community Income

- background grey layer: income choropleth by community area
- point layer: restaurant points by price level (1–4)

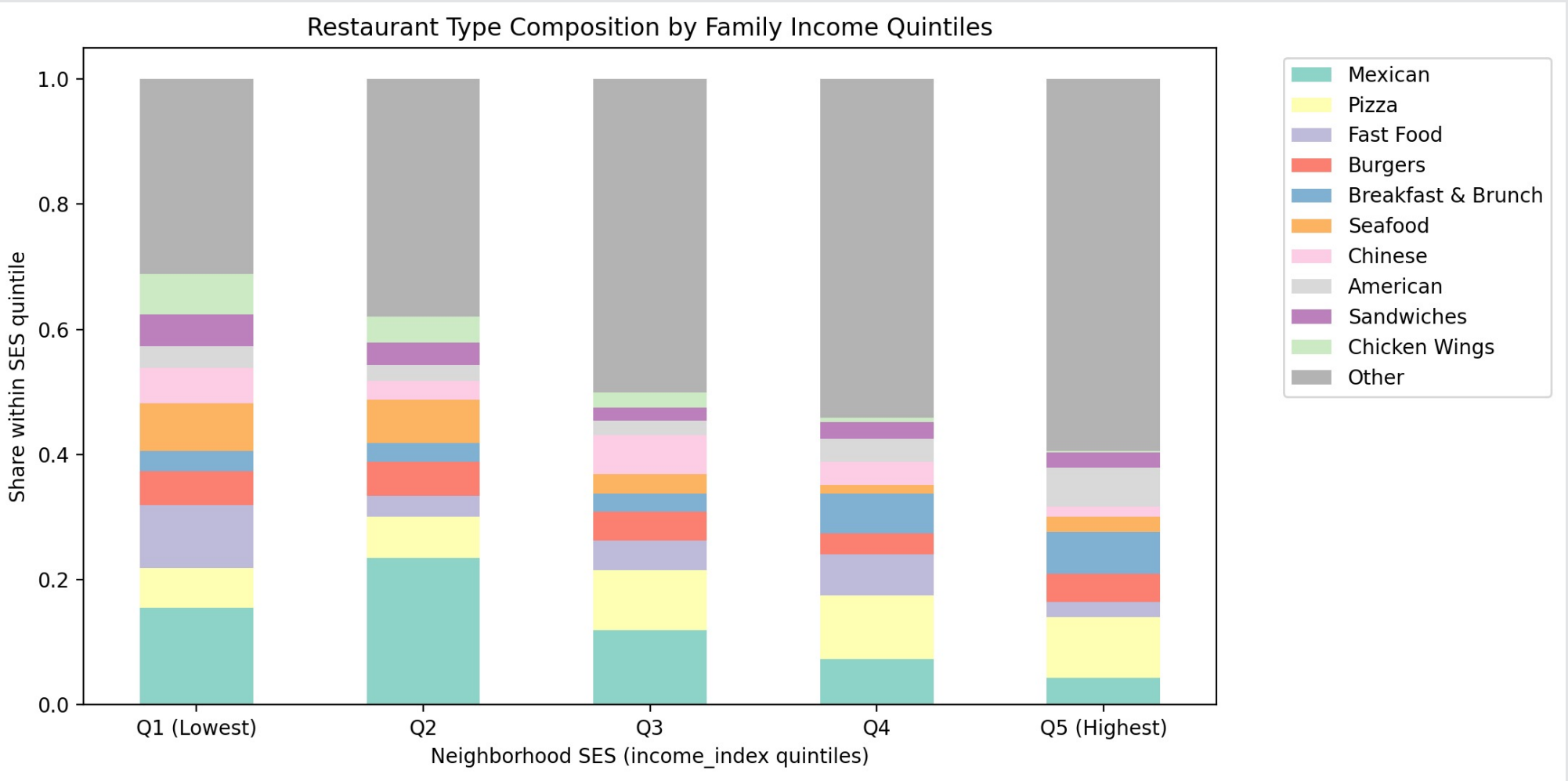
Service Availability Across Family Income Quintiles



Primary Restaurant Types



Restaurant Type Composition by Family Income Quintiles



Conclusions and Limitations

RQ1 (Price–SES):

- Higher-income community areas → higher average price levels
- This relationship remains even after controlling for restaurant type, service features, and rating-based “quality” signals

RQ2 (Characteristics–SES):

- High-priced restaurants and formal dining features (reservable & dine-in) are concentrated in higher-income central and lakefront areas of the city
- Lower-income areas have a higher share of low-tier options and a different cuisine mix

Limitation / next step:

- Future work could incorporate additional SES indicators (e.g., education levels or unemployment rate) to capture neighborhood socioeconomic conditions more comprehensively